

Effect of the KL Coefficient on the Reward–Diversity Trade-off in PPO-based RLHF Fine-tuning: A Case Study on KoChatGPT

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Abstract—The final stage of Reinforcement Learning from Human Feedback (RLHF), PPO fine-tuning, is characterized by a trade-off between reward maximization and a KL penalty that discourages the policy from drifting too far from the initial policy. The KL coefficient β , which controls this trade-off, was left at its untested library default (0.1) in KoChatGPT (airobotlab, 2023; built on ColossalAI Coati), a widely used Korean RLHF tutorial project. We perform a controlled sweep over $\beta \in \{0, 0.02, 0.05, 0.1, 0.3, 1.0\}$, together with a 2-extra-seed replication for three of these conditions ($\beta=0, 0.02, 0.1$), to quantify how β affects reward, empirical KL divergence, and generation diversity in a KoGPT2-based RLHF-PPO pipeline. Our results show that (i) at this experimental scale (1–3 seeds per condition, 10 episodes), all six PPO-fine-tuned conditions achieve a lower mean reward than the SFT baseline; (ii) a sample-level repetition-rate metric reveals repetition-collapse tendencies in the baseline as well as nearly every condition, suggesting that this behavior originates in the SFT stage rather than being induced by β ; (iii) in the seed-repeat experiment, differences across β within a fixed seed are smaller than differences across seeds, indicating that at this scale the effect of β can be masked by seed-level variance; and (iv) a corpus-pooled distinct-n metric can be quantitatively shown to misclassify a condition with genuine per-sample repetition collapse as sufficiently diverse, motivating a sample-level repetition-rate metric as a complementary criterion.

I. INTRODUCTION

RLHF consists of three stages: (i) supervised fine-tuning (SFT), (ii) reward model (RM) training, and (iii) PPO-based policy fine-tuning. In this last stage, the policy is optimized to maximize the reward model’s score while being regularized by a KL divergence term that keeps it from drifting too far from the original (SFT) policy. Although the coefficient β that controls the strength of this regularization is a key RLHF hyperparameter, open-source tutorial implementations often leave it at its untested default value.

We extend **KoChatGPT** [?], [?], a Korean-language tutorial that reproduces an RLHF pipeline for KoGPT2 using the ColossalAI Coati library. Inspecting the actual library source code revealed that the KL coefficient argument (`kl_coef`) of `PPOTrainer` defaults to 0.1, and that the KoChatGPT tutorial uses this value without ever validating it. We extend the original tutorial precisely at this point – by systematically

testing how β actually affects the outcome – through a grid sweep and a multi-seed replication.

A. Research Questions and Hypotheses

- **H1.** Smaller β yields a higher reward-model score but lower generation diversity (a reward–diversity trade-off).
- **H2.** Larger β keeps the policy too close to the SFT model, resulting in little reward improvement.
- **H3.** There exists an intermediate β that improves reward while avoiding severe text collapse.

B. Contributions

- We systematically sweep the previously untested KL coefficient β in the PPO stage of KoChatGPT, and implement the sweep in a reproducible manner grounded in the actual library source (the reward computation and the KL estimator it uses).
- In addition to corpus-pooled distinct-n, we define a sample-level repetition-rate metric and a degeneracy criterion based on it, integrate them into the model-selection rule, and quantitatively show that this metric detects repetition collapse that the corpus-level metric alone misses.
- We replicate three β values with two additional random seeds and quantitatively compare the resulting across- β differences against across-seed variance.

II. BACKGROUND

A. PPO-based RLHF

When fine-tuning an initial policy π_{SFT} (obtained via SFT) against a reward model $r_\phi(x, y)$, the KL-regularized reward is defined as [?], [?]:

$$R(x, y) = r_\phi(x, y) - \beta \cdot D_{\text{KL}}(\pi_\theta(\cdot | x) \| \pi_{\text{SFT}}(\cdot | x)) \quad (1)$$

The policy π_θ is trained with the clipped PPO objective [?]:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2)$$

where $\rho_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ and $\hat{A}_t$ is the GAE-based advantage estimate.

III. RELATED WORK

Base system. KoChatGPT [?], the tutorial we extend, reproduces a KoGPT2-based SFT-RM-PPO pipeline using ColossalAI’s Coati (applications/Chat) [?] library.

KL-regularized RLHF. The objective in Eq. (1) was first proposed by Ziegler et al. [?]; Stiennon et al. [?] established the reward-KL frontier as a standard way to present such results; and Ouyang et al. (InstructGPT) [?] applied it at large commercial scale. Our experimental design (grid sweep, reward-KL frontier plot) follows this lineage directly, and additionally reports confidence intervals from multi-seed replication in a small-scale reproduction setting.

Reward overoptimization. Gao et al. [?] analyzed, via large-scale scaling laws, the phenomenon in which over-optimizing against a reward model raises reward scores without tracking true preferences. The reward model used in our experiments was trained on only a fraction of the available ranking data and may therefore be relatively noisy, which is relevant to the interpretation of results discussed in §VII.

Limitations of diversity metrics. The distinct-n metric was proposed by Li et al. [?]; Holtzman et al. [?] pointed out that corpus-level aggregated metrics can miss degeneration in individual samples. We confirm this limitation numerically in our experiments (§VII).

IV. METHOD

A. Base System

The actor is initialized from the SFT checkpoint, and both the critic and the reward model are initialized from the RM checkpoint; the initial model is a frozen reference policy equal to the SFT policy. PPOTrainer’s default hyperparameters are `kl_coef=0.1`, `eps_clip=0.2`, and `value_clip=0.4`. The reward model was trained on only 10% of the KoChatGPT tutorial’s ranking data (`total_using_ratio=0.1`).

B. Proposed Extension: KL Coefficient Sweep

To quantify how β affects the reward-diversity trade-off, we independently ran PPO fine-tuning over the grid $\mathcal{B} = \{0, 0.02, 0.05, 0.1, 0.3, 1.0\}$ ($\beta=0.1$ corresponds to the tutorial’s untested default). All hyperparameters other than β , and the initial checkpoints, were held fixed so that any observed difference is attributable to β alone. For a fair comparison across conditions, the actor/critic/initial model were re-initialized from the SFT/RM checkpoints for every value of β .

C. Seed-Repeat Experiment

A single-seed sweep alone cannot distinguish whether differences across conditions are a systematic effect of β or noise from the stochasticity of training (rollout sampling, parameter initialization, etc.). To assess this, we repeated training for three conditions ($\beta \in \{0, 0.02, 0.1\}$) with two additional random seeds ($seed \in \{1, 2\}$), and combined these with the original sweep’s result (treated as $seed=0$) to compute

the mean and standard deviation over three observations per condition.

D. Evaluation Protocol

For each condition, we generated with greedy decoding on $N=30$ held-out prompts (not used in training) and measured the following.

- **Reward score:** mean/standard deviation of the score assigned by r_ϕ .
- **Empirical KL:** computed using Schulman’s low-bias k3 estimator [?].

$$\hat{D}_{\text{KL}} = \mathbb{E}[(e^\Delta - 1) - \Delta], \quad \Delta = \log \pi_\theta(a | s) - \log \pi_{\text{SFT}}(a | s) \quad (3)$$

- **Distinct-n [?]:** the unique n-gram ratio computed by pooling all 30 held-out generations together.

$$\text{distinct-}n = \frac{|\{\text{unique } n\text{-grams}\}|}{|\{\text{total } n\text{-grams}\}|} \quad (4)$$

- **Sample-level repetition rate:** measures the degree of repetition within an individual response y , capturing single-sample collapse that a corpus-level distinct-n can miss.

$$\text{rep}(y) = 1 - \frac{|\{\text{unique bigrams in } y\}|}{|\{\text{total bigrams in } y\}|} \quad (5)$$

computed on the generated continuation only (prompt excluded, based on the action mask).

- **Degeneracy flag:** a response y is flagged as degenerate if its repetition rate exceeds a threshold or if the number of generated tokens is too small (i.e., effectively empty).

$$\text{is_degenerate}(y) = \mathbb{I}[\text{rep}(y) > \tau_{\text{rep}} \vee |y| < \tau_{\text{len}}] \quad (6)$$

A condition is marked `any_degenerate=True` if at least one of its 30 evaluation responses is flagged.

E. Model Selection Criterion

The best model is selected as the condition with the highest reward score among those that (a) retain at least 70% of the pre-training (SFT) distinct-2 and (b) have `any_degenerate=False`.

$$\beta^* = \arg\max_{\beta \in \mathcal{B}'} \bar{r}_\phi(\beta), \quad \mathcal{B}' = \{\beta \in \mathcal{B} : \text{distinct2}(\beta) \geq 0.7 \times \text{distinct2}_{\text{SFT}} \wedge \text{any_degenerate} = \text{False}\} \quad (7)$$

If $\mathcal{B}'$ is empty, β^* is reported as undefined (see §VI).

V. EXPERIMENTAL SETUP

VI. RESULTS

All seven conditions are flagged `any_degenerate=True` (i.e., at least one of the 30 evaluation responses satisfies the degeneracy criterion of Eq. (6)), so $\mathcal{B}'$ in Eq. (7) is empty and β^* is undefined (`best_kl_coef=None`). All six PPO conditions achieve a lower mean reward than the SFT baseline (2.545).

As shown in Table III and Fig. 2, the largest difference in mean reward across β is 0.16 ($\beta=0.00$ vs. $\beta=0.10$), whereas

TABLE I: Fixed hyperparameters

Item	Value
Base model	KoGPT2 (skt/kogpt2-base-v2)
PPO data	kochatgpt_3_PPO.jsonl
RM training data fraction	10% (total_using_ratio=0.1)
β grid	{0, 0.02, 0.05, 0.1, 0.3, 1.0}
β values replicated	{0, 0.02, 0.1} (2 extra seeds)
max_epochs	1
train_batch_size	8
num_episodes	10
max_timesteps	3
update_timesteps	3
eps_clip	0.2
value_clip	0.4
Eval. prompts	30 (held-out)
Eval. decoding	greedy
Training decoding	sampling (top-k=50, T=1.0)

TABLE II: β sweep results (single seed per condition, $N=30$)

β	Reward	KL	D-1	D-2	max rep
baseline (SFT)	2.545	0.000	0.683	0.762	0.950
0.00	1.815	0.258	0.407	0.521	0.952
0.02	1.992	0.129	0.608	0.734	0.794
0.05	1.780	0.115	0.652	0.764	0.950
0.10	2.194	0.164	0.449	0.531	0.942
0.30	2.136	0.291	0.494	0.608	0.950
1.00	1.589	0.109	0.495	0.634	0.800

the across-seed standard deviation within each β is 0.18–0.24, i.e., as large as or larger than that. In other words, at this experimental scale, the average difference obtained by changing β is smaller than the difference obtained by simply re-running the same β with a different seed.

VII. DISCUSSION

A. PPO fine-tuning does not exceed the baseline reward

As shown in Table II, all six PPO conditions achieve a lower mean reward (1.59–2.19) than the SFT baseline (2.545). This suggests that (i) the reward model, trained on only 10% of the ranking data, may have had limited generalization, and (ii) the extremely small training budget of 10 episodes may simply have been insufficient for stable policy improvement. Because this pattern appears regardless of β , it is more reasonably attributed to the scale of the experiment than to an effect of β itself.

B. Repetition collapse appears to originate in the SFT stage, not β

The baseline (the SFT model prior to PPO training) already exhibits severe repetition, with $\text{max_repetition}=0.950$. The six PPO-trained conditions also show max_repetition in a similar range (0.794–0.952), suggesting that the observed repetition collapse is not newly induced by PPO or β , but rather a tendency already present from SFT-ing KoGPT2 (125M) on a small amount of data, which PPO simply carries over. That said, $\beta=0.02$ (0.794) and $\beta=1.00$ (0.800) show the relatively lowest max_repetition among the six conditions, a weak signal

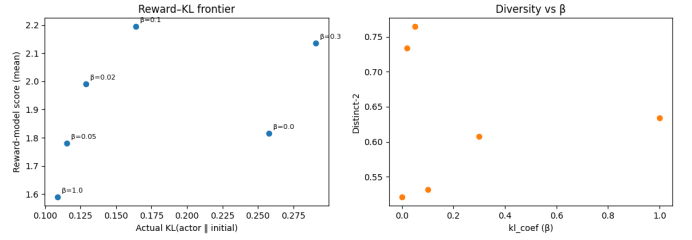


Fig. 1: Reward–KL frontier (left) and distinct-2 as a function of β (right) across the six conditions. Points are shown without connecting lines, reflecting as-is that the empirical KL is not monotonic in the ordering of β .

TABLE III: Seed-repeat experiment: mean $\pm$ std. over 3 seeds ($\beta \in \{0, 0.02, 0.1\}$)

β	Reward	KL	Distinct-2
0.00	2.038 $\pm$ 0.240	0.326 $\pm$ 0.082	0.624 $\pm$ 0.094
0.02	2.102 $\pm$ 0.179	0.284 $\pm$ 0.145	0.695 $\pm$ 0.045
0.10	2.196 $\pm$ 0.181	0.298 $\pm$ 0.120	0.631 $\pm$ 0.088

that repetition may be somewhat mitigated at very small or very large β .

C. The effect of β is within the scale of seed variance

The seed-repeat experiment of §IV-C (Table III) supports this interpretation. Within a fixed seed, changing β across $\{0, 0.02, 0.1\}$ barely changes reward, KL, or distinct-2 (e.g., for seed 2, the reward for $\beta=0.00$ and $\beta=0.02$ was identical to several decimal places), whereas changing the seed produced larger differences. This explains why a reward–KL frontier obtained from a single seed per condition may look different from the smooth curves reported in large-scale studies [?], [?], and suggests that reliably observing the theoretical effect of KL regularization requires multiple seeds per condition and a larger training budget.

D. The blind spot of corpus-level distinct- n

$\beta=0.05$ has a distinct-2 of 0.764, essentially the same as or even higher than the baseline (0.762), comfortably clearing the diversity floor in Eq. (7) ($0.7 \times 0.762 = 0.533$). However, its max_repetition is 0.950, identical to the baseline, meaning at least one of its 30 evaluation responses is still severely repetitive and degenerate. In other words, looking only at the corpus-pooled distinct-2, this condition is incorrectly classified as sufficiently diverse, but combining the sample-level repetition rate (Eq. (5)) and the degeneracy flag (Eq. (6)) reveals that it should in fact be excluded as a candidate. This is a numerically confirmed instance, in our own experiment, of the corpus-level diversity metric limitation pointed out by Holtzman et al. [?], and supports the risk – discussed by Gao et al. [?] – of selecting a model using only reward and diversity metrics derived from a reward model trained on limited data.

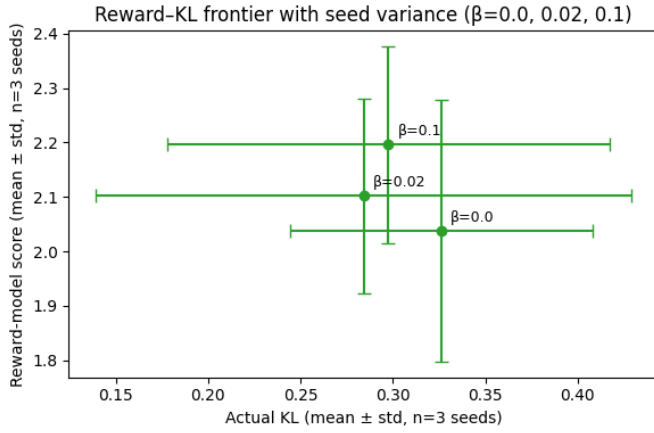


Fig. 2: Reward–KL frontier for $\beta \in \{0, 0.02, 0.1\}$ (mean $\pm$ std., $n=3$ seeds). The error bars are as large as or larger than the differences between condition means, showing that at this scale the effect of β is not clearly distinguishable from seed-level variance.

VIII. LIMITATIONS

This study has the following limitations: (i) the main sweep (6 conditions) was run with a single seed per condition, and seed replication was performed only for three conditions ($\beta \in \{0, 0.02, 0.1\}$, with 2 extra seeds, i.e. 3 observations total); (ii) the training budget is extremely small (10 episodes), so the longer-term effect of β may not be fully observed; (iii) the reward model was trained on only 10% of the ranking data, so the reward signal itself may have limited reliability; (iv) we use a small, 125M-parameter model (KoGPT2), so generalization to larger models needs separate verification; (v) the evaluation set is limited to 30 prompts.

IX. CONCLUSION AND FUTURE WORK

We systematically swept the previously untested KL coefficient β across six conditions in the KoChatGPT PPO tutorial, and verified three of them with additional-seed replication. We found that (i) at this experimental scale, every PPO-trained condition achieves a lower reward than the SFT baseline; (ii) sample-level repetition collapse appears to originate in the SFT stage rather than being caused by β ; (iii) in the seed-repeat experiment, across- β differences are smaller than across-seed differences, indicating that the effect of β can be masked by seed variance at this scale; and (iv) we identified a concrete case where corpus-level distinct-n misses genuine sample-level repetition collapse. Future work includes (1) replicating the full six-condition sweep with multiple seeds to obtain a statistically reliable reward–KL frontier, (2) increasing the fraction of ranking data used to train the reward model, (3) applying the adaptive KL controller proposed by Ziegler et al. [?], and (4) reproducing these experiments with a larger model and training budget.

REFERENCES